GEOMETRY · AREA, SURFACE AREA AND VOLUME

Solids Volume Lab

Explore & Reflect

Name _____ Date _____ Period _____

Open **Solids Volume Lab** on Mr. K's Classroom: Geometry → Area, Surface Area and Volume → Interactive Math. Answer Part 1 **before** you use the tool. Use the tool for Parts 2 and 3, then do Part 4 on your own.

1 Predict

before you touch the tool

1. A cone and a cylinder have the same radius and height. How many cones would fill the cylinder?

_____

_____

2 Explore

use the tool and record what you see

Volume and surface area

Leave π in your answers where it appears.

Solid	Dimensions	Volume	Surface area
Rectangular prism	ℓ = 5, w = 3, h = 4		
Cylinder	r = 3, h = 6		
Cylinder	r = 3, h = 8		
Cone	r = 3, h = 6		
Square pyramid	s = 6, h = 4		
Sphere	r = 3		

Scale the 5 × 3 × 4 prism

k	Volume	Volume ÷ 60	Surface area	SA ÷ 94
1	60	1	94	1
2				
3				

3 Write the rule

fill in each blank from what you observed

1. For a prism or cylinder, V = _____.
2. For a pyramid or cone, V = _____.
3. A cone holds _____ of a cylinder with the same radius and height.
4. For a sphere, V = _____ and SA = _____.
5. Scaling every dimension by k multiplies surface area by _____ and volume by _____.

4 Check no tool: show your work

1. Find the volume of a cylinder with r = 5 and h = 10.

2. Find the volume of a cone with r = 6 and h = 4.

3. Find the volume of a sphere with diameter 12.

4. Two similar solids have a scale factor of 3. Compare their volumes.

5 Reflect

Why does doubling every dimension multiply the volume by 8?